Energy of Simulated Annealing in Pure Spherical Spin Hamiltonians

Upper Bound for the Threshold Energy of Pure Spherical Spin Glasses

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Motivation

Why spherical spin glasses? Spherical p-spin Hamiltonians are canonical models of *rugged random energy landscapes*. They provide a mathematically precise setting for studying high-dimensional nonconvex optimization, algorithmic barriers, and the behavior of noisy dynamics on random objectives, all of which capture features relevant to modern ML optimization.

Why simulated annealing? SA is a natural Langevin-based optimization method: gradually increase the inverse temperature β so the chain gives more weight to high-energy configurations.

Connection to course themes. The analysis uses Langevin-sampling ideas: Markov chains, warm starts, concentration, confinement, and functional inequalities. The soft penalty f_L turns the hard spherical constraint into a radial confining drift, enabling the Lipschitz reduction.

Notation and Preliminaries

Probability space. Consider $(\Omega, \mathbb{P})$ with $\Omega = S_N$ and $\mathbb{P} = \mathrm{Unif}(S_N)$, the rotationally invariant probability measure on the high-dimensional sphere

$$S_N := \sqrt{N}\, S^{N-1} = \{\sigma \in \mathbb{R}^N : \|\sigma\|_2 = \sqrt{N}\}.$$

Partition function, Gibbs measure, and free energy. For inverse temperature $\beta \geq 0$,

$$Z_N(\beta) := \int_{S_N} e^{\beta H_{N,p}(\sigma)}\, d\mathbb{P}(\sigma), \qquad F_N(\beta) := \frac{1}{N} \log Z_N(\beta),$$

$$d\mu_\beta(\sigma) := \frac{e^{\beta H_{N,p}(\sigma)}}{Z_N(\beta)}\, d\mathbb{P}(\sigma), \qquad \mu_0 = \mathbb{P} = \mathrm{Unif}(S_N).$$

Overlap geometry. For $\sigma_1, \sigma_2 \in S_N$, define

$$R(\sigma_1, \sigma_2) := \frac{\langle \sigma_1, \sigma_2 \rangle}{N} \in [-1, 1].$$

Euclidean and spherical gradients. For $x \in S_N$, let

$$P_x^\perp := I_N - \frac{xx^\top}{N}, \qquad \nabla_{\mathrm{sp}} H_{N,p}(x) := P_x^\perp \nabla H_{N,p}(x).$$

Here $P_x^\perp$ projects onto $T_x S_N$, and $\nabla H_{N,p}(x)$ is the Euclidean gradient on $\mathbb{R}^N$.

The Spherical p-Spin Hamiltonian

Spherical Pure p-spin Hamiltonian. For $p \geq 2$, define the random homogeneous polynomial $H_{N,p} : S_N \to \mathbb{R}$ by:

$$H_{N,p}(\sigma) := \frac{1}{N^{(p-1)/2}} \sum_{i_1,\ldots,i_p=1}^{N} g_{i_1\cdots i_p}\, \sigma_{i_1} \cdots \sigma_{i_p}, \qquad \sigma \in S_N,$$

where $g_{i_1\cdots i_p} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. It is a centered Gaussian process with covariance

$$\mathbb{E}\big[H_{N,p}(\sigma_1)H_{N,p}(\sigma_2)\big] = NR(\sigma_1,\sigma_2)^p, \qquad \sigma_1,\sigma_2 \in S_N.$$

Energy density and threshold energy. For $\sigma \in S_N$, the normalized energy is

$$E_N(\sigma) := \frac{H_{N,p}(\sigma)}{N}.$$

The optimization problem is to find configurations with large energy density:

$$\max_{\sigma \in S_N} E_N(\sigma).$$

For the pure model, $E_\infty(p) = 2\sqrt{\frac{p-1}{p}}$ marks the accessible high-energy edge of the p-spin landscape, beyond which Lipschitz-hardness creates an algorithmic barrier.

Spherical Langevin Dynamics and Simulated Annealing

Spherical Langevin dynamics. At inverse temperature β, the dynamics on S_N is

$$d\sigma_t = \left(\beta \nabla_{\mathrm{sp}} H_{N,p}(\sigma_t) - \frac{N-1}{2N}\sigma_t\right) dt + P^{\perp}_{\sigma_t}\, dB_t.$$

Here $P^{\perp}_{\sigma_t}$ projects the Brownian noise onto $T_{\sigma_t} S_N$.

Simulated annealing runs the same dynamics with an increasing inverse-temperature schedule

$$0 = \beta_0 < \beta_1 < \cdots < \beta_K = \beta^*.$$

1. **Initialize:** $x_0 \sim \mathrm{Unif}(S_N)$.
2. **Stage** k**:** run Langevin at β_k, warm-started from x_{k-1}.
3. **Output:** x_K after the temperature ramp.

Pure benchmark and motivation. For pure p-spin, $E_\infty(p)$ is the Langevin threshold and the Lipschitz-hardness benchmark. This talk focuses only on the *upper bound* via soft spherical approximation and Lipschitz hardness.

Statement of the Upper Bound

Theorem ([Sel24], Theorem 1.7)
For any $p \geq 2$ and $\beta_0 < \infty$, there exists $\delta = \delta(p, \beta_0) > 0$ such that for all $\beta \leq \beta_0$ and all $T \geq 0$, if $\sigma_0 \in S_N$ is independent of $H_{N,p}$, then for N sufficiently large, for some $c = c(p, \beta_0, T) > 0$,

$$\mathbb{P}\left[\sup_{t \in [0,T]} \frac{H_{N,p}(\sigma_t)}{N} \leq E_\infty(p) - \delta\right] \geq 1 - e^{-cN}.$$

Interpretation. Langevin dynamics remains below the threshold energy $E_\infty(p)$ with exponentially high probability.

Proof I: Soft Spherical Approximation

Why introduce a soft process? The spherical process $\sigma_t \in S_N$ has two complications: (i) the Brownian noise is projected onto the moving tangent space $T_{\sigma_t} S_N$; and (ii) the process evolves on a curved manifold rather than in a fixed Euclidean space.

Soft spherical replacement. Replace the hard constraint $\sigma_t \in S_N$ by a full-space process $y_t^{(L)} \in \mathbb{R}^N$ with radial confinement:

$$dy_t^{(L)} = \left(\beta_t \nabla H_{N,p}(y_t^{(L)}) - f_L'\left(N^{-1}\|y_t^{(L)}\|^2\right) y_t^{(L)}\right) dt + dB_t,$$

where $f_L(r) = L(r-1)^2 + (r^2-1)^p$. Normalize back to the sphere:

$$\widetilde{y}_t^{(L)} := \frac{\sqrt{N}\, y_t^{(L)}}{\|y_t^{(L)}\|}.$$

Soft approximation lemma
For L large enough,

$$\mathbb{P}\left[\sup_{t\in[0,T]} \|\sigma_t - \widetilde{y}_t^{(L)}\| \le \varepsilon\sqrt{N}\right] \ge 1 - e^{-cN}.$$

This mirrors a Langevin-sampling idea: replace a hard geometric constraint by a confining drift keeping the diffusion near $\|y\|^2/N \approx 1$.

Proof II: Lipschitz Hardness

Step 1: Lipschitz approximation. For fixed Brownian noise, the soft endpoint $y_T^{(L)}$ agrees w.h.p. with a dimension-free Lipschitz map of the disorder:

$$H_{N,p} \mapsto \ell_B^{(L)}(H_{N,p}).$$

Grönwall stability of the soft SDE gives the Lipschitz dependence.

Step 2: Lipschitz hardness. No dimension-free $O(1)$-Lipschitz algorithm exceeds $E_\infty(p)$ by a constant amount w.h.p. [HS25]:

$$\mathbb{P}\left[\frac{H_{N,p}(\ell_B^{(L)}(H_{N,p}))}{N} \geq E_\infty(p) + \eta\right] \leq e^{-cN}.$$

Step 3: Transfer and refinement. Soft approximation + C-boundedness first gives $E_\infty(p) + \eta$. A gradient lower bound plus a short restart improves this to

$$\sup_{t\in[0,T]} \frac{H_{N,p}(\sigma_t)}{N} \leq E_\infty(p) - \delta \quad \text{w.h.p.}$$

Sampling perspective: The soft penalty f_L replaces the hard spherical constraint by radial confinement, enabling the Lipschitz reduction.

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